

36. Let the numbers be $3x$ and $4x$. Then, their H.C.F. = x . So, $x = 4$.
So, the numbers are 12 and 16.
L.C.M. of 12 and 16 = 48.
37. Let the required numbers be $27a$ and $27b$.
Then, $27a + 27b = 216 \Rightarrow a + b = 8$.
Now, co-primes with sum 8 are (1, 7) and (3, 5).
 $\therefore$ Required numbers are $(27 \times 1, 27 \times 7)$ and $(27 \times 3, 27 \times 5)$ i.e., (27, 189) and (81, 135).
Out of these, the one available in the given alternatives is the pair (27, 189).
38. Let the required numbers be $33a$ and $33b$.
Then, $33a + 33b = 528 \Rightarrow a + b = 16$.
Now, co-primes with sum 16 are (1, 15), (3, 13), (5, 11) and (7, 9).
 $\therefore$ Required numbers are $(33 \times 1, 33 \times 15)$, $(33 \times 3, 33 \times 13)$, $(33 \times 5, 33 \times 11)$, $(33 \times 7, 33 \times 9)$.
The number of such pairs is 4.
39. Numbers with H.C.F. 15 must contain 15 as a factor.
Now, multiples of 15 between 40 and 100 are 45, 60, 75 and 90.
 $\therefore$ Number-pairs with H.C.F. 15 are (45, 60), (45, 75), (60, 75) and (75, 90).
[$\because$ H.C.F. of (60, 90) is 30 and that of (45, 90) is 45]
Clearly, there are 4 such pairs.
40. Out of the given numbers, the two with H.C.F. 12 and difference 12 are 84 and 96.
41. Let the numbers be $37a$ and $37b$.
Then, $37a \times 37b = 4107 \Rightarrow ab = 3$.
Now, co-primes with product 3 are (1, 3).
So, the required numbers are $(37 \times 1, 37 \times 3)$ i.e., (1, 111).
 $\therefore$ Greater number = 111.
42. Let the numbers be $13a$ and $13b$.
Then, $13a \times 13b = 2028 \Rightarrow ab = 12$.
Now, co-primes with product 12 are (1, 12) and (3, 4).
So, the required numbers are $(13 \times 1, 13 \times 12)$ and $(13 \times 3, 13 \times 4)$.
Clearly, there are 2 such pairs.
43. Since the numbers are co-prime, they contain only 1 as the common factor.
Also, the given two products have the middle number in common.
So, middle number = H.C.F. of 551 and 1073 = 29;
First number = $\left(\frac{551}{29}\right) = 19$;
Third number = $\left(\frac{1073}{29}\right) = 37$.
 $\therefore$ Required sum = $(19 + 29 + 37) = 85$.
44. Let the numbers be $13x$ and $15x$.
Then, their L.C.M. = $195x$.
So, $195x = 39780$ or $x = 204$.
 $\therefore$ The numbers are 2652 and 3060.
45. Let the numbers be $3x$, $4x$ and $5x$.
Then, their L.C.M. = $60x$.

So, $60x = 2400$ or $x = 40$.

$\therefore$ The numbers are (3×40) , (4×40) and (5×40) .

Hence, required H.C.F. = 40.

46. Let the numbers be $2x$, $3x$, $5x$ and $7x$ respectively.

Then, their L.C.M. = $(2 \times 3 \times 5 \times 7)x = 210x$.

[$\because$ 2, 3, 5, 7 are prime numbers]

So, $210x = 630$ or $x = 3$.

$\therefore$ The numbers are 6, 9, 15 and 21.

Required difference = $21 - 6 = 15$.

47. Other number = $\left(\frac{12 \times 336}{84}\right) = 48$.

48. L.C.M. = $\frac{324}{3} = 108$.

49. Product of numbers = H.C.F. $\times$ L.C.M.

$$\Rightarrow pq = x \times \text{L.C.M.}$$

$$\Rightarrow \text{L.C.M.} = \frac{pq}{x} = \frac{p(xy)}{x} = py.$$

50. Let the numbers be x and $(2000 - x)$.

Then, their L.C.M. = $x(2000 - x)$.

$$\text{So, } x(2000 - x) = 21879$$

$$\Leftrightarrow x^2 - 2000x + 21879 = 0$$

$$\Leftrightarrow (x - 1989)(x - 11) = 0$$

$$\Leftrightarrow x = 1989 \text{ or } x = 11.$$

Hence, the numbers are 1989 and 11.

51. Let the numbers be x and $4x$.

$$\text{Then, } x \times 4x = 84 \times 21 \Leftrightarrow x^2 = \left(\frac{84 \times 21}{4}\right) \Leftrightarrow x = 21.$$

Hence, larger number = $4x = 84$.

52. Let the numbers be x and $(100 - x)$.

$$\text{Then, } x(100 - x) = 5 \times 495$$

$$\Leftrightarrow x^2 - 100x + 2475 = 0$$

$$\Leftrightarrow (x - 55)(x - 45) = 0$$

$$\Leftrightarrow x = 55 \text{ or } x = 45.$$

$\therefore$ The numbers are 45 and 55.

Required difference = $(55 - 45) = 10$.

53. Let the numbers be x and $(x + 2)$.

$$\text{Then, } x(x + 2) = 24$$

$$\Leftrightarrow x^2 + 2x - 24 = 0$$

$$\Leftrightarrow (x - 4)(x + 6) = 0 \Leftrightarrow x = 4.$$

So, the numbers are 4 and 6.

54. Let the numbers be a and b .

$$\text{Then, } a + b = 36 \text{ and } ab = 3 \times 105 = 315.$$

$$\therefore \text{ Required sum} = \frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab} = \frac{36}{315} = \frac{4}{35}.$$

55. Let H.C.F. be h and L.C.M. be l .

$$\text{Then, } l = 12h \text{ and } l + h = 403.$$

$$\therefore 12h + h = 403 \text{ or } h = 31.$$

$$\text{So } l = (403 - 31) = 372.$$

$$\text{Hence, other number} = \left(\frac{31 \times 372}{93}\right) = 124.$$